

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: FPGA Based System Design (01CT0718)	Aim: To design and implement digital code conversion systems for efficient data format transformation using FPGA.	
Experiment No: 6	Date:	Enrolment No: 92301733013

Aim: To design and implement digital code conversion systems for efficient data format transformation using FPGA.

Software required: Xilinx Vivado

Hardware required: Nexys4 DDR FPGA Development Board

Theory:

A binary-to-gray code converter is a type of code converter that can translate a binary code into its equivalent gray code.

The binary-to-gray code converter accepts a binary number as input and produces a corresponding gray code as output.

Truth Table:

Natural-binary code				Gray code			
B3	B2	B1	B0	G3	G2	G1	G0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

Boolean Expression:

$$g_0 = b_0b_1' + b_1b_0' = b_0 \oplus b_1$$

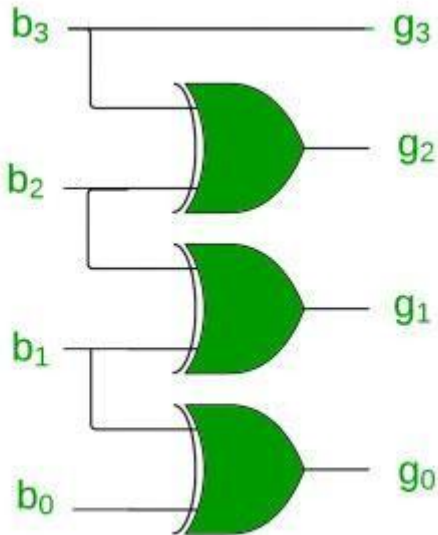
$$g_1 = b_2b_1' + b_1b_2' = b_1 \oplus b_2$$

$$g_2 = b_2b_3' + b_3b_2' = b_2 \oplus b_3$$

$$g_3 = b_3$$

Circuit Diagram:

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: FPGA Based System Design (01CT0718)	Aim: To design and implement digital code conversion systems for efficient data format transformation using FPGA.	
Experiment No: 6	Date: _____	Enrolment No: 92301733013



Verilog Code:

Binary to gray

```

module binary_to_gray(
    input [3:0] B,
    output [3:0] G
);
    assign G[3] = B[3];
    assign G[2] = B[3] ^ B[2];
    assign G[1] = B[2] ^ B[1];
    assign G[0] = B[1] ^ B[0];
endmodule

```

gray to binary

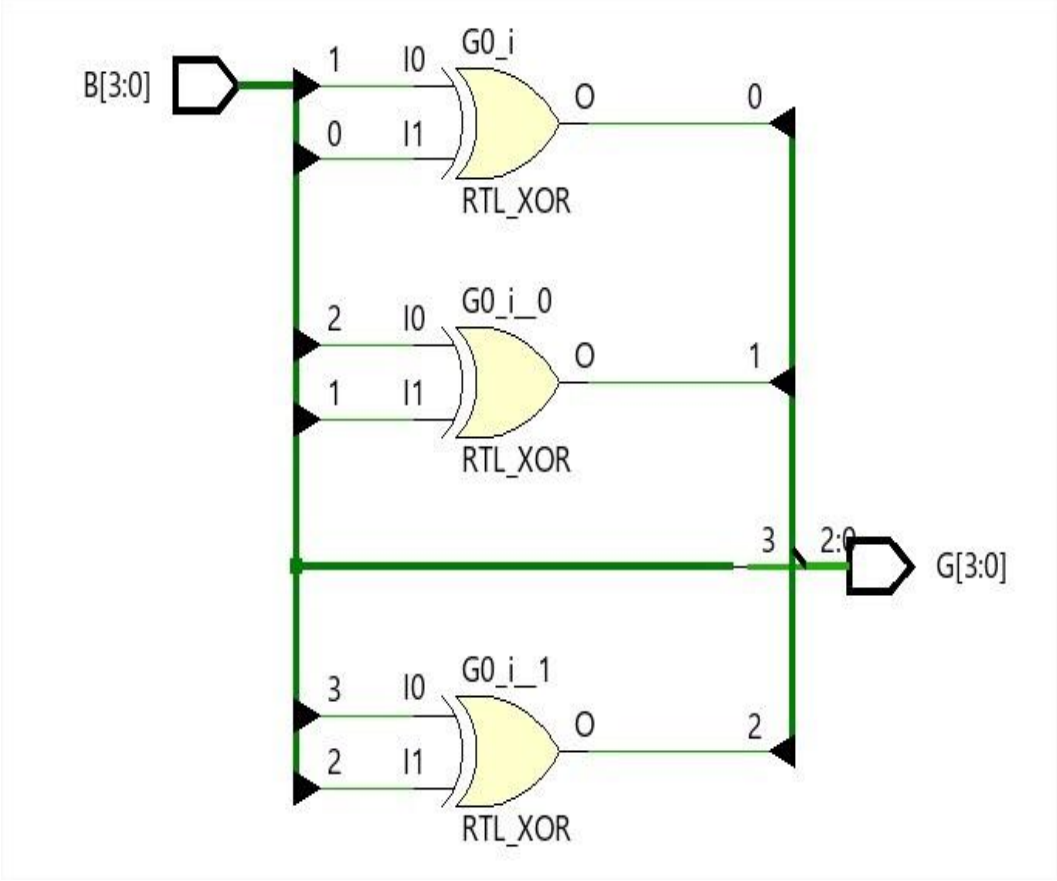
```

module gray_to_binary(
    input [3:0] G,
    output [3:0] B
);
    assign B[3] = G[3];
    assign B[2] = B[3] ^ G[2];
    assign B[1] = B[2] ^ G[1];
    assign B[0] = B[1] ^ G[0];
endmodule

```

Simulation Results:

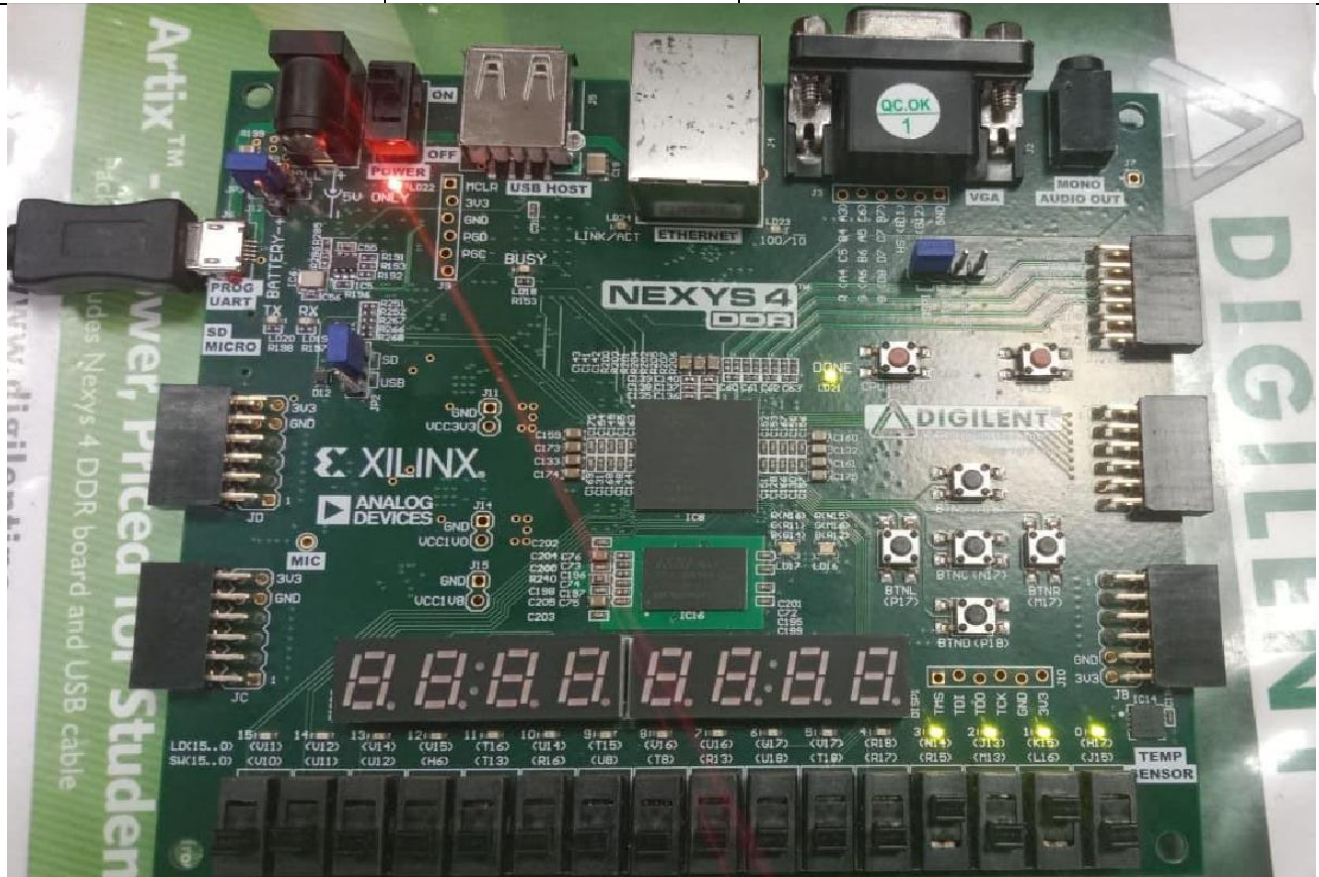
 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: FPGA Based System Design (01CT0718)	Aim: To design and implement digital code conversion systems for efficient data format transformation using FPGA.	
Experiment No: 6	Date:	Enrolment No: 92301733013



		160.000 ns																
	Name	Value	0.000 ns					50.000 ns					100.000 ns					150.000 ns
			0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	
>	 B[3:0]	f	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
>	 G[3:0]	8	0	1	3	2	6	7	5	4	c	d	f	e	a	b	9	8



Aim: To design and implement digital code conversion systems for efficient data format transformation using FPGA.

Enrolment No: 92301733013

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: FPGA Based System Design (01CT0718)	Aim: To design and implement digital code conversion systems for efficient data format transformation using FPGA.	
Experiment No: 6	Date:	Enrolment No: 92301733013

Conclusion:

The binary-to-gray code converter was successfully designed, simulated, and implemented on the FPGA development board using Verilog HDL. The simulation and hardware outputs matched the expected truth table, confirming the correct conversion of binary inputs to Gray code. This experiment demonstrated the practical implementation of code converters on an FPGA and highlighted the importance of simulation and hardware verification in digital system design.

Post Lab Exercise:

1. Why are FPGA-based code converters preferred over discrete logic implementations in modern digital systems?

FPGA-based code converters are widely preferred because they offer flexibility, faster development, lower cost, and reduced hardware requirements. Since they are reconfigurable, the design can be easily modified by updating the HDL code without making any changes to the physical circuit.

2. Why should invalid input combinations be explicitly handled in code converter designs?

Invalid input combinations should be properly handled to avoid incorrect or unpredictable outputs. This helps improve the overall reliability of the system and ensures that the circuit operates safely and correctly under different input conditions.

3. Why are code converters used in digital systems?

Code converters are used to convert data from one binary code format to another, enabling communication and compatibility between different digital devices and systems.

4. What is the advantage of gray code over binary code?

Gray code changes only one bit at a time between consecutive numbers, which reduces transition errors and glitches. This makes it ideal for position encoders and digital communication systems.

5. What is the purpose of assigning FPGA pins in Quartus?

FPGA pin assignment in Quartus connects the design's input and output signals to the appropriate physical pins on the FPGA. This allows the implemented circuit to interact properly with external components such as switches, LEDs, buttons, and other devices.

6. How can the output of a code converter be verified on the FPGA board?

The output can be tested by applying various input combinations through switches or buttons and observing the resulting outputs on LEDs or seven-segment displays. The observed results are then compared with the expected values in the truth table to confirm that the circuit is functioning correctly.